Giant Killing in Football: An Information-Theoretic Analysis of Underdog Upsets Using Causal Emergence

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Abstract

Underdog wins are a beloved part of any sport, and are a fundamental part of the history of soccer, where many giant killings take place with historic wins such as Senegal beating the World Cup reigning champions France in their first World Cup match ever in 2002, Greece winning the 2004 UEFA European Championship, and Leicester City winning the 2015/16 Premier League season. Even as these games are beloved by fans, statistically interpreting and predicting why and when an underdog team wins is generally opaque. We test whether they carry a distinct emergent coordination, defined by the causal-emergence coefficient Ψ of Rosas et al., computed on tracking data reconstructed from StatsBomb 360 freeze frames for 136 matches (30 upsets, 106 rank-matched controls) across eight competition-seasons, 2020–2025. We extend on prior work on causal emergence in football in four ways through non-parametric Kraskov–Stogbauer–Grassberger mutual-information estimation instead of a Gaussian estimation, interaction of orders $\ell = 1, 2, 3$ rather than $\ell = 1$ alone, and inter-team coupling metrics.

The results show that the level of Ψ does not sufficiently determine underdog wins. None of 54 pre-specified mixed-effects tests reached $p < 0.05$ before correction, the order profile does not differ by outcome ($p = 0.995$ for the Match Type $\times$ Order interaction), the favourited team shows no measurable disruption, and a random forest on 260 trajectory features classifies upset versus control at chance (ROC-AUC 0.547, permutation $p = 0.26$). However, the trajectory does seem to signal an underodg win. The slope over match time of the underdog-minus-favourite emergence gap is positive in controls and negative in upsets (Hedges' $g = -0.47$, permutation $p = 0.024$, 6 of 18 feature $\times$ order cells survive false-discovery-rate correction). Because negative Ψ denotes redundant, strictly organized play, this suggests that underdogs in giant killings become progressively more structured relative to their opponents as the match develops. This is a temporal signature, and the opposite of the level-based prediction we set out to test.

We also show that the whole-minus-sum criterion is a poor instrument in this experiment, and that its failure is specific to macro features that are linear aggregates of the micro state. For those features $\Psi > 0$ in 0.004 % of windows there is no indication for the hypothesised positive emergence, while for velocity-similarity clustering, which is not a linear aggregate, $\Psi > 0$ in 25.3 %. Across the experiment Ψ is almost entirely determined by the micro sum ($r = 0.995$). A further 24 % of individual mutual-information terms sit at the estimator's ceiling of 2.57 nats, but we trace 88.6 % of that saturation to a preprocessing defect — a double-counted match clock that opens a 42-minute interpolated void in every match — rather than to the estimator; outside the void saturation is 3.8 %. The gap-trend result survives excluding those windows ($g = -0.458$, $p = 0.009$). Minimum detectable effect at 80 % power is $|g| \approx 0.58$. We report the nulls, the one live effect, and the measurement pathology together, and give concrete recommendations through a ΦID-based synergy measure, non-linear macro features, longer windows, and a pre-registered replication of the gap-trend endpoint.

Keywords: causal emergence; partial information decomposition; football analytics; underdog; collective behaviour; null results.

Contents

1 Introduction

Giant killings are some of the most celebrated events in football. Leicester City winning the Premier League after having been promoted from a league below, Senegal over France in 2002, Saudi Arabia over Argentina in 2022, and more recently with the 2026 World Cup in teams such as Cabo Verde. These results are captivating because they defy expectation. Football is, on the available evidence, more pron to underdog wins, where across twelve team ball sports, soccer has one of the highest empirical rates of a historically weaker side beating a stronger one [18], a finding consistent with earlier work on the parity and predictability of competitions [1].

Yet the explanations offered are almost entirely post hoc and qualitative, unable to be expressed quantitatively, where the underdog seems to have "wanted it more" or the giant became complacent or luck simply intervened. None are factually quantifiable explanations. A coach cannot train "wanting it more" into a team, and an analyst cannot quantify complacency. The literature on quantitative football analytics has also not been able to predict underdog wins quantitatively either. Its dominant instruments rely on expected goals, possession, passing networks, physical output which describe what a team did rather than how it organised itself to do it [10, 3]. They are, moreover, largely reductive since they aggregate individual contributions and therefore cannot in principle detect a team property that is not the sum of its players.

Causal emergence helps to discover this discrepancy. Rosas et al. [14], building on partial information decomposition [19] and on earlier accounts [6, 15], give a formal criterion for when a macroscopic feature of a multivariate system carries predictive information about the system's future that no individual part carries on its own. Applied to football, the thus question becomes whether a team's collective configuration in its centre of mass, compactness, interaction network predicts its own future in a way that the individual player positions do not? Cheng, Chang and Doya [4] answered this affirmatively for ordinary match dynamics, showing on 34 J-League matches that average causal emergence over positional macro features correlates strongly with ball possession ($R = 0.74$–0.79) and rises for the attacking side in the sixty seconds before a shot.

We seek to build on top of Cheng, Chand, and Doya by asking if giant killings reflect a genuine, transient surge of collective organisation in the weaker side, then underdogs in upsets should show atypically high emergence, favourites should show disrupted or collapsed emergence, or both. The intuition follows the idea that as the game progresses, favourites become complacent in defence and can allow a counterattack to become successful. We set out to test this with four extensions to [4], with a non-parametric mutual-information estimator that does not assume joint Gaussianity, higher interaction orders, so that coordination localised in pairs or triples of players such as a centre-back pairing, a midfield triangle and can be distinguished from team-wide coordination, explicit inter-team coupling, so that disruption of the favourite can be measured rather than inferred, and a pre-specified analysis plan with matched controls, correction for multiplicity, and a sensitivity battery.

We found that the level-based hypothesis fails under every robustness variant we examined. What survives is a temporal effect that runs opposite to the direction we predicted. Relative to the favorites, underdogs in upsets become progressively more redundancy-dominated, more drilled, and less synergistic as the match proceeds, while in control matches the reverse holds. Underneath both results sits a measurement problem we did not anticipate and that we regard as the most transferable contribution of this paper. The whole-minus-sum criterion Ψ, evaluated on 60-second windows of football tracking data, is dominated by definitional redundancy. A second, more contingent problem sits alongside it: a double-counted match clock filled 29 % of every match with interpolated motion, which none of our diagnostics detected and which accounts for the great majority of the estimator saturation we initially attributed to sample size. We report both, and what each does and does not affect.

Contributions. (i) The first application of causal emergence to match outcomes rather than in-match phases on a list of upsets and controls. (ii) A null Ψ level that does not distinguish giant killings, at any order, for any macro feature, under any sensitivity variant, and does not support classification above chance. (iii) One exploratory positive result which is the emergence gap trend. (iv) An explanation of why the whole-minus-sum criterion is the wrong instrument, with quantitative evidence via the redundancy decomposition, together with a diagnosed pre-processing defect that inflated the apparent estimator ceiling, and specific alternatives. (v) A complete, open replication package.

Section 2 sets out the theoretical framework. Section 3 describes the dataset, the tracking reconstruction, the estimator and the analysis plan. Section 4 reports results by research question. Section 5 interprets them, draws out the practical implications, and states what would settle the question.

2 Background

2.1 Causal emergence

Consider a system of n parts observed at times t and $t' > t$, with micro state $X_t = (X_t^1, \ldots, X_t^n)$, and a resulting feature V_t that adds no predictive power over $X_{t'}$ once X_t is known exactly, so that $V_t - X_t - X_{t'}$ is a Markov chain. Rosas et al. [14] define V_t to exhibit causal emergence of order k when its unique information about the future is positive in the partial information decomposition sense.

$$\mathrm{Un}^{(k)}(V_t; X_{t'} \mid X_t) > 0. \tag{1}$$

Computing (1) requires a choice of PID functional and is generally intractable. Rosas et al. therefore supply a practical sufficient criterion that depends only on pairwise mutual informations

$$\Psi_{t,t'}^{(k)}(V) \;:=\; I(V_t; V_{t'}) \;-\; \sum_{|\alpha|=k} I(X_t^{\alpha}; V_{t'}) \tag{2}$$

where α ranges over subsets of parts of size k and X_t^{α} is their joint state. $\Psi^{(k)} > 0$ is sufficient but not necessary for causal emergence.

The interpretation used throughout this paper follows [14, 4]. $\Psi > 0$ is synergy-dominated where the collective carries predictive information that the parts do not. $\Psi < 0$ is redundancy-dominated where the parts, taken together, predict the macro future better than the macro state does, which is the signature of synchronised, mutually predictable, well-drilled behaviour. Rosas et al. are explicit that (2) belongs to the whole-minus-sum family and therefore double-counts redundancy across all n marginals, so that redundancy drives Ψ negative and a negative value is inconclusive rather than evidence of absence. This caveat, stated in their §III A, turns out to govern our results so we return to it in section 4.1.

2.2 Causal emergence in football

Cheng et al. [4] instantiate this framework on football tracking data using four macroscopic features via the team centre of mass V_{CoM}, the mean distance of players to the field centre V_{dist}, and two weighted clustering coefficients on the player interaction graph, one with edge weight $w_{uv} = 1/d_{uv}$ (inverse distance, $V_{\mathrm{clust,dist}}$) and one with $w_{uv} = S_u S_v(\cos \Delta\theta + 1)$ (velocity similarity, $V_{\mathrm{clust,vel}}$), computed with the Onnela definition [12]. Their micro features are the corresponding per-player quantities. Restricting to $\ell = 1$ and estimating mutual information with a Gaussian plug-in, they find that match-average Ψ on V_{dist} and V_{CoM} correlates strongly with the home–away possession differential ($R = 0.79$ and 0.74), that $V_{\text{clust,dist}}$ correlates moderately ($R = 0.58$), and that $V_{\text{clust,vel}}$ does not ($R = 0.06$). In the sixty seconds before a shot, the attacking team's Ψ rises and the defending team's falls.

Two features of that design matter for ours. First, their tracking data are proprietary 25 Hz optical recordings, down-sampled to 1 Hz which are accurately complete and detailed. Second, their analysis falls within matches because it compares phases and roles, not outcomes across matches. Both differences bear directly on our results.

2.3 Upsets

Existing quantitative work on upsets is largely descriptive and event-based. Analyses of World Cup upsets find that winning underdogs concede possession, passing accuracy and touches while producing more shots on target, clearances and aerial duels. Vicente et al. [18] approach the phenomenon structurally, ranking sports by an underdog achievement score and attributing football's high score to randomness factors intrinsic to the game (goal size relative to the number of defenders, low scoring frequency, ball handling). Wunderlich et al. [20] show that the randomness contribution to goal scoring decreases as a match progresses, which disadvantages weaker teams.

None of this literature addresses collective organisation. Our question is whether the tactical profile the event data describe has an information-theoretic correlate at the level of team structure.

2.4 Research questions

RQ1 (Emergence signatures). Do giant killings show different Ψ from expected results such as underdogs higher, favourites lower?

RQ2 (Order). At what interaction order does any such effect manifest: team-wide ($\ell = 1$), pairs ($\ell = 2$), or triples ($\ell = 3$)?

RQ3 (Temporal dynamics). Does underdog emergence build gradually or in bursts? Is there a tipping point at which the underdog's Ψ overtakes the favourite's?

RQ4 (Disruption). Can we quantify the underdog disrupting the favourite's coordination?

RQ5 (Spatial signatures). Which macro features, if any, discriminate giant killings?

We additionally defined three summary metrics: the Emergence Asymmetry Index EAI $= (\Psi_u - \Psi_f)/(\Psi_u + \Psi_f + \epsilon)$, the Disruption Coefficient DC $= I(V_u; V_f)/H(V_f)$, and the Synergy–Redundancy Ratio SRR $= \max(0, \Psi)/(|\min(0, \Psi)| + \epsilon)$. All three decline on continuous football data, and section 3.5 documents the failure modes and the stabilised replacements we use.

3 Data and methods

3.1 Data

Upset definition. A match is a candidate giant killing when the pre-match mismatch is large and the weaker side wins. We compute a continuous upset score in $[0, 1]$ from five normalised pre-match gaps which are rolling Elo including a 70-point home advantage (weight 0.40), log squad-value ratio from Transfermarkt (0.25), form differential over the previous ten matches (0.20), pre-match expected-goals differential from Understat (0.10), and a manager-tenure proxy

(0.05), all multiplied by 1.0 if the underdog won, 0.5 for a draw and 0.0 for a loss, and thresholded at 0.65. The threshold was set so that roughly 10 % of fixtures in a typical top-five league season are flagged, matching the empirical shock-result frequency.

Controls. For each flagged upset we retain rank matched fixtures meeting the same mismatch criteria in which the favourite won as expected. This yields a matched-mismatch design where every match in the dataset, upset or control, is a fixture with a large paper gap. This is deliberate and it has a consequence we flag now and return to in section 4.5 since the dataset contains no even-game layer, so a general losing effect cannot be separated from a giant-killing specific one.

Dataset. Intersecting the upset and control lists with matches for which StatsBomb 360 freeze frames are available [17] gives 136 matches: 30 upsets and 106 controls, spanning 2020-10-17 to 2025-07-22 (table 1). The dataset is dominated by international tournaments, which is a generalisability limitation: club football, and in particular the cup competitions where giant killings are most frequent, is barely represented, only having matches from specific teams and seasons such as Barcelona 20/21 league matches and Bayer Leverkusen 23/24 league matches.

Competition	**Matches**	**Upsets**	**Window**
FIFA World Cup 2022	40	11	Nov–Dec 2022
FIFA Women's World Cup 2023	37	8	Jul–Aug 2023
UEFA Euro 2024	24	6	Jun–Jul 2024
UEFA Women's Euro 2022	12	1	Jul 2022
UEFA Women's Euro 2025	10	2	Jul 2025
UEFA Euro 2020	7	1	Jun 2021
Bundesliga 2023/24	3	0	Oct 2023–May 2024
La Liga 2020/21	3	1	Oct 2020–Feb 2021
Total	**136**	**30**	2020-10-17 – 2025-07-22

Table 1: Upsets and controls. The design is matched-mismatch. Controls are fixtures with the same pre-match gap in which the favourite won.

3.2 Tracking reconstruction

StatsBomb 360 provides a freeze frame of visible player positions at each on-ball event, roughly one frame every 1.9 s, concentrated near the ball, with off-camera players absent. We reconstruct continuous 1 Hz trajectories following Penn et al. [13] utilizing an ARMAX one-step predictor for player position with the ball position as exogenous input, Hungarian assignment of observed positions to trajectories under that predictive model, and velocity-corrected interpolation between observed points, in which the linear interpolant is corrected by the time integral of the visible mean team velocity scaled by $\alpha \approx 0.5$.

Penn et al. report a mean error of 7.33 m for off-camera players on held-out data, against 3.45 m overall. The LSTM–GNN imputer of Everett et al. [5] achieves ~ 6.9 m from event data alone and is the natural cross-validation benchmark; we did not run it, and this remains the most important unaddressed check on our input data. Reconstruction error is symmetric and partially averages out over a 60 s window, but it is not innocuous for the clustering coefficients, which are non-linear in the pairwise distances.

For preprocessing, positions are resampled to a 1 Hz grid; velocities are finite differences; goalkeepers are dropped and each team is truncated at its first red card, so that all analyses use exactly ten outfielders per side, following [4]. The tracking clock is aligned to the event clock as $t = (\text{period} - 1) \cdot 2700 + 60 \cdot \text{minute} + \text{second}$. Windows are 60 s long and advance in 10 s steps.

3.3 Features

Six macro features are computed per team per window: the four of [4] plus two composites.

Name	Dim	Definition
`V_com`	2	centre of mass, $\frac{1}{P}\sum_i (x_i, y_i)$
`V_dist`	1	mean distance to pitch centre, $\frac{1}{P}\sum_i \lVert p_i - (60, 40) \rVert$
`V_cdist`	1	Onnela weighted clustering coefficient, $w_{uv} = 1/d_{uv}$
`V_cvel`	1	Onnela weighted clustering coefficient, $w_{uv} = S_u S_v(\cos\Delta\theta + 1)$
`V_all`	5	all four jointly
`V_fast`	3	`V_com` + `V_dist`

Table 2: Macroscopic features. `V_com` and `V_dist` are linear aggregates of the micro state, the clustering coefficients are not.

Micro features are the $2P = 20$ individual coordinates. At $\ell = 1$ the sum in (2) has 10 terms, at $\ell = 2$ it has $\binom{10}{2} = 45$, and at $\ell = 3$ it has $\binom{10}{3} = 120$. At $\ell = 3$ we retain only triples whose members are pairwise within 25 m of one another in mean window position, capped at 60 triples, and rescale the sum by $n_{\text{total}}/n_{\text{selected}}$. This Horvitz–Thompson-style correction assumes the retained triples are representative, which proximity selection deliberately violates. Both the scaled and unscaled sums are emitted, and the induced bias is measured against the exhaustive sum.

3.4 Estimation

All information quantities use the Kraskov–Stögbauer–Grassberger estimator #1 [9]. For samples $\{(x_i, y_i)\}_{i=1}^{N}$ and neighbour count k, with ϵ_i twice the distance to the kth nearest neighbour in the joint space under the Chebyshev norm and $n_x^{(i)}, n_y^{(i)}$ the marginal counts within ϵ_i,

$$\hat{I}(X;Y) = \psi(k) - \frac{1}{N}\sum_{i=1}^{N}\Big[\psi\Big(n_x^{(i)} + 1\Big) + \psi\Big(n_y^{(i)} + 1\Big)\Big] + \psi(N). \tag{3}$$

Differential entropies for the Disruption Coefficient use the Kozachenko–Leonenko estimator [8]. Production runs use $k = 4$; $k \in \{3, 4, 7, 10\}$ is varied in sensitivity analysis.

Equation (3) was validated against the closed form for a bivariate Gaussian, $I = -\frac{1}{2}\ln(1 - \rho^2)$, with absolute error below 10^{-2} nats at $N = 4000$, $k = 4$ across $\rho \in \{0, 0.3, 0.6, 0.9\}$, and the entropy estimator against $\frac{d}{2}\log(2\pi e)$ for $d \in \{1, 2, 3, 5\}$. An $O(N^2)$ transcription test and the k-d-tree implementation agree with the implementation.

Three estimator defects, are found and corrected. These are reported because each would have produced a plausible-looking but wrong result. (i) Ties produced infinities. Because the 1 Hz grid interpolates linearly between sparse events, the velocity-derived $V_{\text{clust,vel}}$ is piecewise constant over long runs with an absolute neighbour tolerance those runs give $\epsilon_i = 0$, a neighbour count of -1, and $\psi(0) = -\infty$. The fix is a relative tolerance, clamping of counts at zero, and tie-breaking jitter of 10^{-10} in standardised units, as Kraskov et al. recommend. (ii) Chebyshev scale dominance. KSG uses the max norm so a joint feature vector is governed by whichever dimension has the largest numerical spread. Raw macro features span $\sigma(V_{\text{CoM}}) \approx 19\,\text{m}$ against

$\sigma(V_{\text{clust,dist}}) \approx 0.10$; left unstandardised, the five-dimensional `V_all` returned numerically identical Ψ to the three-dimensional `V_fast`, the clustering dimensions contributing nothing. All estimates therefore use per-window z-scored inputs with V_t and $V_{t'}$ sharing one scaler. Mutual information is invariant to this transform in theory; the estimator is not. (iii) Cost. Per-subset tree builds are pure overhead at $N \approx 59$; the micro-side distances do not depend on the macro feature. Precomputing the $2P$ per-coordinate distance matrices once per window and forming subset distances by elementwise maximum gives bit-identical results at a fraction of the cost.

3.5 The three proposed metrics, and their stabilised forms

Each metric is emitted exactly as specified and in a repaired form, because all three degenerate here.

EAI. $(\Psi_u - \Psi_f)/(\Psi_u + \Psi_f + \epsilon)$ is bounded in $[-1, 1]$ only when both Ψ are non-negative. Empirically $\Psi < 0$ almost everywhere so the denominator is a sum of two same sign large numbers that pass through zero exactly when the teams are comparable which is where the hypothesis locates the interesting behaviour. The stabilised form uses $|\Psi_u| + |\Psi_f| + \epsilon$ and is bounded, same sign, same monotonicity, and no pole.

DC. $I(V_u; V_f)/H(V_f)$ divides by a differential entropy which is scale-dependent and here routinely negative (≈ -5 nats), negating the "fraction of entropy explained" explanation. We store $I(V_u; V_f)$ and $H(V_f)$ separately and report the informational coefficient of correlation $\text{dc}_{ic} = 1 - e^{-2I}$, bounded in $[0, 1)$ and invariant to smooth reparameterisation of V.

SRR. $\max(0, \Psi)/(|\min(0, \Psi)| + \epsilon)$ evaluated at a single Ψ collapses to 0 or to $\Psi/\epsilon \approx 10^9$ and is a ratio statistic requiring a set of values. Per-window, rolling-block and whole-match versions are emitted. On the linear macro features Ψ is essentially never positive, so match-level SRR is zero or near-zero throughout and is itself a finding (section 4.1).

3.6 Analysis plan

We implement window thinning where Ψ is emitted every 10 s from a 60 s window so that consecutive records share 50 of 60 seconds. All models therefore use non-overlapping windows only (window_start mod 60 = 0), retaining 231,492 of 1,384,566 records (16.7 %), with residual within-match dependence absorbed by a random intercept.

For the variance structure, the intraclass correlation of Ψ is between 0.004 and 0.014 depending on macro feature. 98–99 % of the variance is window to window within a single match. A between-group difference of ~ 0.2 nats sits inside a within-match standard deviation of ~ 3.2 nats, which is why pooled dataset medians appear identical between groups and why the random intercept is mandatory rather than decorative.

For the model, RQ1 fits, for all 6 macro features $\times$ 3 orders,

$$\Psi_{ijt} = \beta_0 + \beta_1 \text{Upset}_{ij} + \beta_2 \text{Time}_t + \beta_3 (\text{Upset} \times \text{Time})_{ijt} + \beta_4 \text{Role}_{ij} + \beta_5 (\text{Upset} \times \text{Role})_{ij} + u_i + \epsilon_{ijt}, \tag{4}$$

with $u_i \sim \mathcal{N}(0, \sigma_u^2)$ a per-match random intercept and time and role centred, so β_1 estimates the average upset effect rather than its value at kickoff for the favourite. Coefficients are reported as $\beta_1/\hat{\sigma}_u$ to permit comparison across orders, since $|\Psi|$ scales mechanically with the number of subsets. All 54 tests form one family under Benjamini–Hochberg [2]. Because upsets end with the underdog ahead by construction, goal times were extracted from the event feed (405 goals across 136 matches), each window carries the goal difference at its start, and every RQ1 model is refitted with that covariate.

RQ2 is Order (3, within) $\times$ Match Type (2, between) $\times$ Role (2, within), match as subject, fitted as a linear mixed model so sphericity need not be assumed, with Greenhouse–Geisser-corrected F-tests reported alongside. Ψ is z-scored within order before testing, otherwise the Order main effect is reflecting only that $120 > 45 > 10$ subsets are being summed. RQ3 compares mean Ψ curves on a normalised-time grid by permutation over whole match labels, locates change points in the gap $g(t) = \Psi_{\text{und}}(t) - \Psi_{\text{fav}}(t)$ by binary segmentation with an L_2 cost [7], and compares per-match least-squares slopes of $g(t)$. RQ4 fits favourite-only models and the disruption coefficient. RQ5 trains a random forest on 260 match-level trajectory summaries with stratified 5-fold cross-validation, class-weight balancing, a 200-draw permutation null in which the whole CV pipeline is re-run on shuffled labels, and temporal blocking.

In determining sensitivity, alternative upset definitions (rank gap $\geq 0, 10, 20, 30$), MAD outlier exclusion, effective-sample thresholds, excluding the first and last 10 % of each match, window lengths $W \in \{30, 60, 90\}$ s and $k \in \{3, 4, 7, 10\}$ recomputed from scratch on a 60-match subsample, a 324-cell specification curve [16] over {summary statistic} $\times$ {windowing} $\times$ {feature} $\times$ {order} $\times$ {role}, and a placebo calibration repeating the entire specification curve on 25 label shuffles.

We deviated from the plan slightly since (i) The proposal targeted 50–75 upsets and the 360 dataset delivered 30, (ii) RQ4's third condition that the favourite loses to an equally strong opponent does not exist in a matched-mismatch dataset and was not tested, and (iii) The gap trend family analysed in section 4.4 was examined after the pre-specified tests returned null. It is exploratory and is reported as such.

4 Results

The production run processed all 136 matches, producing 1,384,566 window-level records over 116,709 distinct windows, with 99.98 % of estimates computed successfully and a median of 59 usable frames per 60 s window. Reconciliation against an independent earlier implementation is accurate on 9,371 audited windows in which the recomputed coefficient matches the stored value to 0.0 nats.

4.1 The measure behaves pathologically before any hypothesis is tested

Ψ is negative in 95.5 % of windows, and this is structural rather than empirical (fig. 1a). Across the dataset, the micro sum averages 12.3 nats against 1.4 for the macro term, a factor of 8.5, so that $\Psi \approx -\sum_i I(X_t^i; V_{t'})$ with $\text{corr}(\Psi, -\sum_i I) = 0.9996$ (fig. 1b). Two mechanisms produce this. First, whole-minus-sum measures double-count redundancy across all n marginals, as Rosas et al. note [14]. Second, V_{CoM} is the mean of the micro positions, so every player is mechanically informative about it, in every match and every team, regardless of tactics.

Moreover, the degeneracy is confined to macro features that are linear aggregates of the micro state (table 3). For V_{CoM}, V_{dist}, `V_fast` and `V_all`, Ψ exceeds zero in 0.004 % of windows pooled and in no window at all for `V_fast`. The Synergy–Redundancy Ratio, defined as a ratio of positive to negative parts, is correspondingly zero for all 136 matches on `V_fast` and for all but a handful on the others. For the two clustering coefficients, which are not linear in the micro state, the picture is different: $\Psi > 0$ in 2.0 % of windows for $V_{\text{clust,dist}}$ and 25.3 % for $V_{\text{clust,vel}}$, and match-level SRR is non-zero in all 136 matches for both, though still small in absolute terms (max = 0.14), so even these features remain redundancy-dominated. The same ordering shows up in the medians where $|\Psi|$ is largest for the linear aggregates (V_{CoM}: -15.4, `V_fast`: -14.1) and smallest for the clustering coefficients ($V_{\text{clust,vel}}$: -2.4, $V_{\text{clust,dist}}$: -5.3).

Macro feature	linear?	median $\Psi^{(1)}$	% windows $\Psi > 0$	matches with SRR > 0
`V_com`	yes	-15.39	0.003	2 / 136
`V_fast`	yes	-14.10	0.000	0 / 136
`V_all`	mixed	-10.52	0.005	7 / 136
`V_dist`	yes	-9.20	0.008	8 / 136
`V_cdist`	no	-5.33	1.99	136 / 136
`V_cvel`	no	-2.42	25.26	136 / 136

Table 3: Headroom for positive emergence, by macro feature, over the 231,492 non-overlapping analysis windows. Where the macro feature is a linear aggregate of the micro state, the criterion has effectively no positive range, and where it is not, it does.

At $N \approx 59$ samples per window and $k = 4$, the KSG estimator has a finite-sample ceiling where no per-player term exceeds 2.5652 nats, and 24 % of all per-player mutual-information terms sit at that ceiling (fig. 1c). Because $\Psi^{(1)} = I_{\text{macro}} - \sum_i I$, ten saturated terms impose a floor at $\Psi \approx -23.1$ nats, and 29–34 % of window-level Ψ values for V_{CoM}, V_{dist} and `V_fast` lie below -23 nats.

The saturation is largely a preprocessing artefact, not an intrinsic property of the estimator. We report this because it materially qualifies the paragraph above. The absolute match clock was computed as $t = (\text{period} - 1) \cdot 2700 + 60 \cdot \text{minute} + \text{second}$, but StatsBomb's `minute` field already runs continuously across the match — period 2 spans minutes 45–92 — so the formula double-counts the first half and displaces every second-half event 45 minutes into the future. This opens a median 2,516 s ($\approx$ 42 min) void between the end of the first half and the shifted start of the second. The 1 Hz reconstruction interpolates linearly across it for any player observed in both halves, so those windows are filled with perfectly smooth synthetic motion while reporting $n_{\text{eff}} = 59$; none of the six diagnostics in section 4.1 detects them. Such motion makes every player maximally informative about the centre of mass, which is exactly the condition that drives the micro sum to the estimator's ceiling.

The arithmetic is stark. Windows overlapping the void are 29.1 % of the analysis panel. Within them 71.7 % of per-player terms are saturated; outside them the figure is 3.8 %. 88.6 % **of all saturation in the corpus lies inside the artefact**, and mean $\Psi^{(1)}$ is -17.0 inside against -8.4 outside. The honest statement is therefore that the KSG estimator does have a ceiling at 2.5652 nats at this sample size, but it is reached almost only when the input is degenerate — and here the degeneracy was manufactured by the clock bug rather than by $N \approx 59$.

Two things are unaffected. The definitional-redundancy floor of the preceding paragraphs is arithmetic, not artefact: V_{CoM} is the mean of the micro positions in every window, void or not. And the one live result survives: recomputing the gap-trend endpoint with void-overlapping windows excluded gives $g = -0.458$ ($p = 0.009$) against the published $g = -0.468$ ($p = 0.005$). The level-based nulls are likewise unlikely to move, since the artefact is close to identical across groups and roles and so differences out. A full re-estimation on the corrected clock is nevertheless the right next step, and the replication pipeline exposes it as `--clock corrected`; every number in this paper was produced under the legacy clock and should be read with that caveat.

The consequence for the hypothesis is immediate. We predicted that underdogs would show positive emergence in bursts. On the four macro features that dominate the analysis by magnitude, and on which [4] reported its strongest results, there is no headroom in which such bursts could be observed at all. The prediction is not so much refuted as untestable with this instrument on these features.

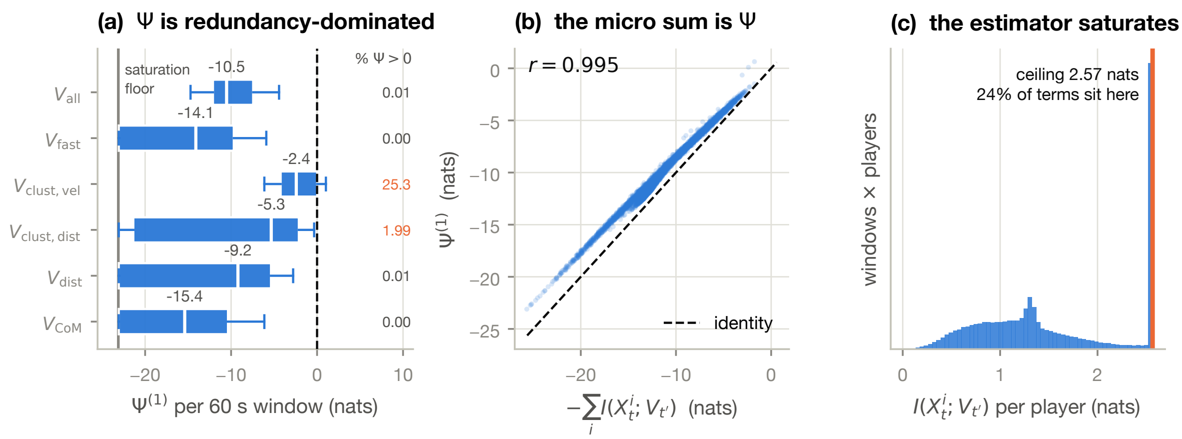

Figure 1: The criterion is dominated by redundancy and is estimator-saturated. (a) Distribution of $\Psi^{(1)}$ per 60 s window by macro feature (box: quartiles, whiskers: 5th–95th percentile, annotation: median, right column: percentage of windows with $\Psi > 0$). $|\Psi|$ is largest, and positive range scarcest, for the linear aggregates, the two non-linear clustering coefficients are the only features with usable headroom. The grey line marks the saturation floor at -23.1 nats. (b) Ψ against the negated micro sum for `V_all` (6,000-window sample), $r = 0.995$. The vertical offset from the identity line is $I(V_t; V_{t'})$, the entire macro contribution. (c) Distribution of the per-player term $I(X_t^i; V_{t'})$ for the linear features: 24 % of terms sit at the estimator's ceiling of 2.57 nats at $N \approx 59$, $k = 4$. Note that 88.6 % of that saturation falls inside the clock-artefact void described in the text; outside it the figure is 3.8 %.

4.2 RQ1 — the level of Ψ carries no outcome signal

Null. Zero of 54 coefficient tests from (4) reach $p < 0.05$ before correction (fig. 2); the joint likelihood-ratio test dropping all upset terms is non-significant for every feature–order pair (smallest $p = 0.18$, all $p_{\mathrm{FDR}} = 0.95$). Adjusting for goal difference does not change this. The largest standardised effect anywhere is $V_{\mathrm{clust,vel}}$ at $\ell = 1$ ($\beta_1/\hat{\sigma}_u = -0.43$, $p = 0.081$) — notably, the one macro feature that is not an average of its micro inputs.

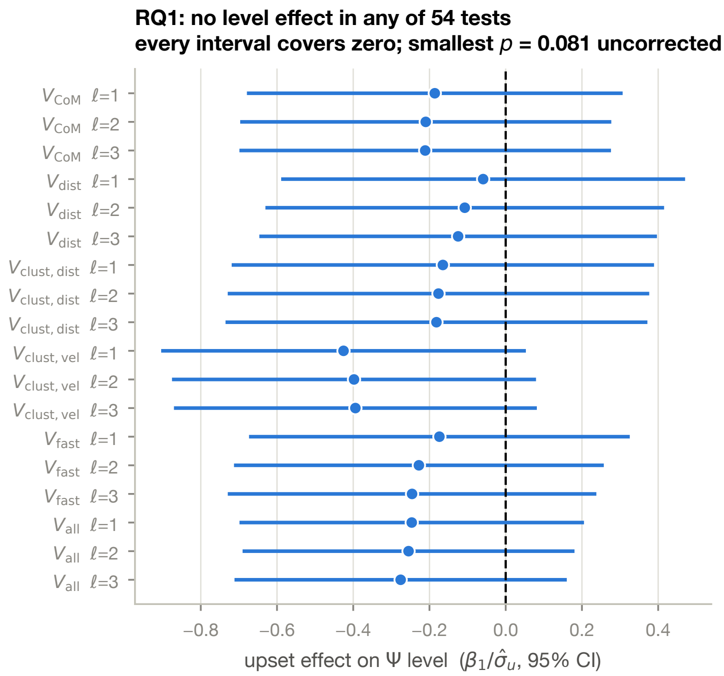

Figure 2: RQ1 is null. Standardised upset effect on the level of Ψ from the pre-specified mixed model (4), for each macro feature × interaction order, with 95 % confidence intervals. Every interval covers zero. The most negative point estimate is $V_{\text{clust,vel}}$ at $\ell = 1$.

4.3 RQ2 — no order-specific signature

Null. The RQ2 question is the Match Type × Order interaction, at $\chi^2 = 0.010$, $p = 0.995$, partial $\eta^2 < 0.001$. There is no evidence that upset emergence manifests preferentially at pair or triple level. The Match Type main effect is also null ($p = 0.188$). The only significant term is the Role main effect ($\chi^2 = 24.09$, $p < 10^{-6}$, partial $\eta^2 = 0.029$), which merely restates that underdogs and favourites differ is true by construction in a mismatch dataset. That the Order main effect is non-significant ($p = 0.844$) confirms the within-order standardisation removed the mechanical scaling as intended. Per-cell repeated-measures ANOVAs with Greenhouse–Geisser correction agree (all $p_{\text{GG}} > 0.34$; largest partial $\eta^2 = 0.031$, upset underdogs).

The methodological extension to $\ell = 2, 3$ — one of the four ways this work was to improve on [4] — therefore bought no additional discrimination. Whether that is a fact about football or about the criterion (section 4.1) cannot be settled from these data.

4.4 RQ3 — levels are null but gap trend is not

Levels and tipping points are null. Mean Ψ curves on a normalised-time grid are indistinguishable between groups under permutation over whole match labels (preserving within-match autocorrelation), for both $\sup_t |\Delta(t)|$ and $\int \Delta^2$ statistics; nothing survives correction (smallest $p_{\text{FDR}} = 0.54$). Neither the change-point location in $g(t)$ ($p = 0.76$), nor the first zero-crossing ($p = 0.27$), nor the fraction of the match with $g > 0$ ($p = 0.50$) separates the groups. In most matches g never crosses zero at all, so there is no tipping point to predict. The RQ3 hypothesis as originally posed is not merely unsupported but inapplicable.

Effects in the gap trend. Fitting a least-squares line to $g(t)$ over normalised match time and comparing per-match slopes between groups gives the one live result in this study (fig. 3, fig. 4, table 4). Nine of 18 feature–order cells are significant raw and six survive Benjamini–Hochberg, with $|g|$ from 0.34 to 0.62 at or above the dataset minimum detectable effect. Control slopes are positive in all 18 cells. A permutation test on the primary specification (`V_all`, $\ell = 1$, 20,000 draws shuffling whole match labels) gives an observed slope difference of -0.489 nats per match, $p = 0.024$.

Feature	Order	upset slope	control slope	g	p	p_{FDR}
`V_all`	3	-3.706	$+4.678$	-0.560	0.0001	0.0015
`V_all`	2	-1.311	$+1.677$	-0.550	0.0002	0.0015
`V_fast`	3	-1.916	$+4.797$	-0.383	0.0009	0.0056
`V_all`	1	-0.207	$+0.282$	-0.468	0.0046	0.0175
`V_fast`	2	-0.432	$+1.734$	-0.340	0.0049	0.0175
`V_cdist`	2	-2.418	$+1.330$	-0.624	0.0117	0.0350

Table 4: The six feature–order cells surviving FDR correction on the gap-slope comparison. Slopes are in nats per normalised match. Control slopes are positive in all 18 cells, including the twelve not shown.

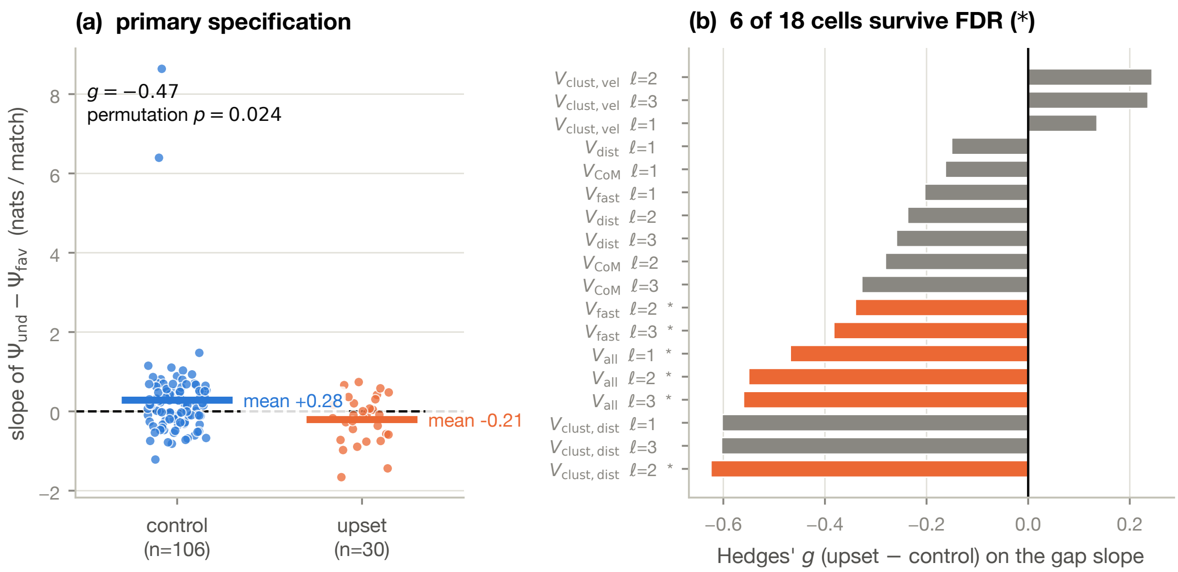

Figure 3: The emergence gap trends down in upsets and up in controls. (a) Per-match least-squares slope of $g(t) = \Psi_{\mathrm{und}}(t) - \Psi_{\mathrm{fav}}(t)$ for the primary specification (`V_all`, $\ell = 1$), horizontal bars are group means. (b) Hedges' g for the upset–control slope difference in each of the 18 feature $\times$ order cells, starred cells survive FDR correction. $V_{\mathrm{clust,vel}}$ is the only feature with the opposite sign, and is not significant in any cell.

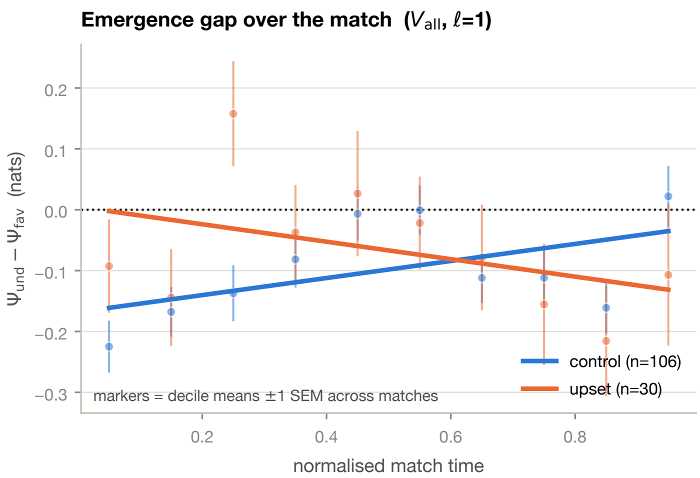

Figure 4: The same effect as a trajectory. Decile means of the emergence gap over normalised match time, ±1 SEM across matches, with the fitted trend for each group. The two groups start indistinguishable and diverge. Note that the mean-curve comparison itself is *not* significant (section 4.4), the tested quantity is the distribution of per-match slopes shown in fig. 3, not the difference between these two curves.

Interpretation. More negative Ψ is redundancy-dominated, which [14, 4] gloss as synchronised, well-drilled, structured play. A downward gap slope in upsets means that the underdog becomes progressively more drilled relative to the favourite as the match develops, while in controls the reverse holds. The downward slope is a trajectory, not a static level, which is why section 4.2 missed it. Its direction is opposite to the RQ1 expectation that underdogs would display higher, positive emergence.

Robustness. Two control matches have visibly extreme positive slopes. Removing them strengthens the result ($g = -0.67$, $p = 0.0064$), winsorising at the 5th/95th percentiles also strengthens it ($g = -0.62$, $p = 0.0049$), a permutation test on the median difference gives $p = 0.012$. Group medians (-0.128 upset, $+0.201$ control) carry the same sign as the means, and the primary test is based on rank so the effect is not an artefact of tail behaviour.

Caveats. This family was examined after the pre-specified tests returned null, so it is exploratory and requires out-of-sample confirmation. $V_{\text{clust,vel}}$ shows the opposite (non-significant) sign in all three orders, so the effect is not uniform across macro features. Because the $\ell = 2$ and $\ell = 3$ cells are near-deterministic rescalings of $\ell = 1$ within a feature, "6 of 18" overstates the number of independent replications so the effective count is closer to three, one per feature family.

4.5 RQ4 — no measurable disruption

The proposal specified three conditions: favourite wins; favourite loses to the underdog; favourite loses to an equally strong opponent. The third does not exist in this dataset (section 3.1). A general losing effect therefore cannot be separated from a giant-killing specific one. This is a design limitation, not an analysis choice, and it means RQ4 as posed is unanswerable with these data.

On the two available conditions, neither the favourite's Ψ (smallest $p = 0.263$, $V_{\text{clust,vel}}$) nor the disruption coefficient dc_{ic} (smallest $p = 0.397$) differs by outcome, nothing survives correction.

Cross-team coupling is high and near-identical between groups ($\mathrm{dc}_{ic} \approx 0.92$), which is consistent with both teams tracking the same ball and suggests dc_{ic} is measuring the shared ball trajectory rather than tactical disruption.

4.6 RQ5 — classification at chance

Each match contributes seven summary statistics per (macro feature × order × role) trajectory plus EAI and DC summaries: 260 features for 136 matches, a feature-to-observation ratio of 1.9 which is what the overfitting regime the design anticipated. Under stratified 5-fold cross-validation the random forest achieves ROC-AUC 0.547 and PR-AUC 0.279 against a base rate of 0.221; the permutation null, in which the entire pipeline is re-run on 200 label shuffles, has mean 0.490 and standard deviation 0.083, giving $p = 0.255$ (fig. 5a). Out-of-time performance under temporal blocking is unstable across split points (0.71, 0.60, 0.54 as the training window grows), which with 7–12 upsets per test block is sampling noise, not evidence of a usable model.

Feature importance. Standard per-feature permutation importance returns exactly zero for all 260 predictors. This is the consequence of extreme redundancy among near-duplicate columns (`psi_l1_mean` versus `psi_l1_median` on the same trajectory), so permuting one leaves the forest able to reconstruct it from its neighbours. Importance is therefore computed out-of-fold, the drop in cross-validated AUC, with permutation applied to held-out rows only, and grouped, with all 50 columns belonging to one macro feature permuted together, which is also the form in which RQ5 poses the question (fig. 5b).

Only $V_{\text{clust,dist}}$ (+0.036) and `V_all` (+0.027) have positive importance, both within roughly 1.5 standard deviations of zero, and four blocks have negative importance, permuting them improves out-of-fold AUC, which is what the noise looks like. Since the underlying model does not generalise, no block should be described as discriminating giant killings. RQ5 has no answer beyond none at this sample size.

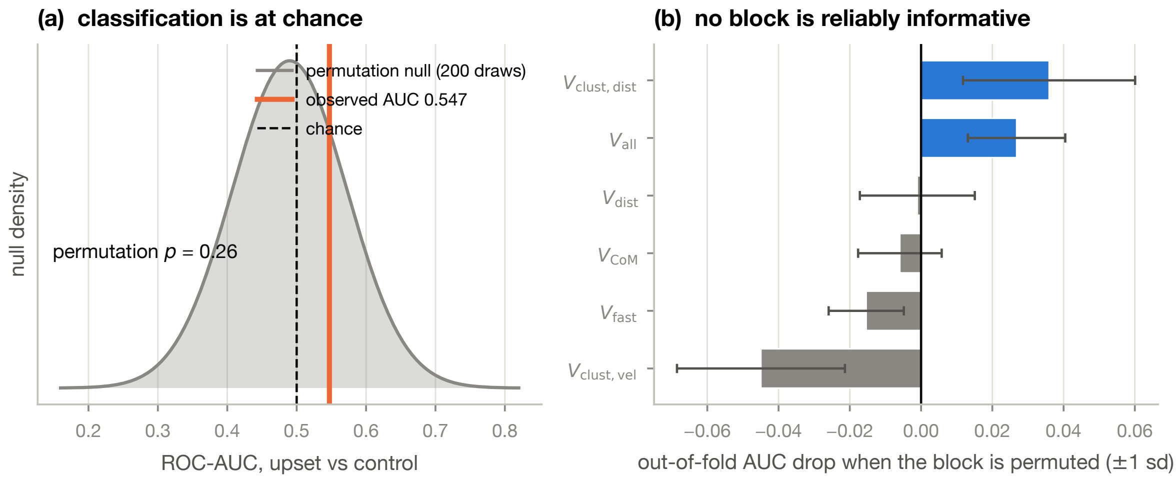

Figure 5: RQ5 is at chance. (a) Observed cross-validated ROC-AUC against the 200-draw permutation null (drawn as a normal approximation with the null's mean and standard deviation). (b) Grouped out-of-fold importance: the AUC drop when all 50 columns of one macro-feature block are permuted together, ±1 sd across folds. Four of six blocks have negative importance.

4.7 Sensitivity

Specification curve. An earlier informal pass appeared to find $p < 0.05$ effects and used the median as the within-match summary of Ψ. Nothing in the data justifies that over the mean, with skewness 0.03, excess kurtosis -0.03 and yet on identical data we have median $p = 0.020$, 10 % trimmed mean $p = 0.058$, mean $p = 0.120$. Rather than choose, we enumerate all 324 combinations (fig. 6a). Median-based specifications clear $p < 0.05$ about twice as often as mean-based ones (30.6 % versus 14.8 %) so that single choice does real work, since mean and trimmed-mean specifications supply over half the significant cells. 323 of 324 specifications give $g < 0$ and the single exception is $+0.00003$. The direction is effectively unanimous even where the magnitude is not significant.

Placebo calibration. Repeating the entire specification curve on 25 label shuffles gives a placebo mean of 3.6 % of cells at $p < 0.05$ (sd 8.5 %) against the observed 21.9 %, an empirical $p = 0.080$ (fig. 6b). The real labels produce roughly six times the placebo rate. We read this as weak evidence for a small real effect, while noting it does not license any single specification as a headline.

Definition, outliers, window and k. The upset coefficient is small and non-significant under every variant of the upset definition (rank gap $\geq 0, 10, 20, 30$), after dropping MAD-flagged outliers, after requiring $n_{\text{eff}} \geq 40$, and after excluding the first and last 10 % of each match. Recomputing Ψ from scratch at $W \in \{30, 60, 90\}$ s and $k \in \{3, 4, 7, 10\}$ on a 60-match subsample shifts $|\Psi|$ substantially, as expected, but leaves the `V_all` upset effect near zero throughout. Notably, the $V_{\text{clust,vel}}$ standardised effect is stable between -0.48 and -0.72 across every W and k, and reaches nominal significance at $W = 90$ ($p = 0.027$) and at $k = 10$ ($p = 0.032$) which is consistent with a small, real, consistently-signed effect on the one non-linear macro feature, and consistent with the reading that longer windows relieve the saturation of section 4.1.

Power. The dataset resolves only $|g| \gtrsim 0.58$ at 80 % power (fig. 6c). Power at $g = 0.3$ is 0.31; at $g = 0.5$ it is 0.68. The proposal asked for 50–75 matches per group; the upset arm delivered 30. Reaching 80 % power at $g = 0.3$ against 106 controls would require roughly 150 upsets.

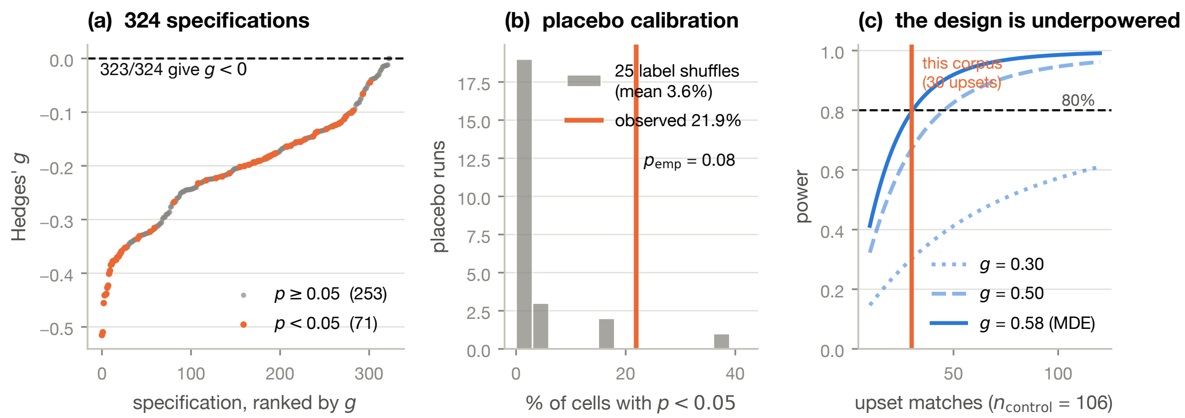

Figure 6: Sensitivity. (a) Specification curve: 324 analytic specifications ranked by effect size; 323 give $g < 0$. (b) Placebo calibration: the observed rate of $p < 0.05$ cells against 25 label shuffles of the same curve. (c) Power against the number of upset matches, holding controls at 106, for three effect sizes.

4.8 Summary of results

RQ	Question	Tests	Verdict
RQ1	emergence signatures	54	no effect detected
RQ2	order of emergence	7	order profile does not differ by outcome
RQ3.a	levels, tipping point	17	no tipping point; curves indistinguishable
RQ3.b	gap trend	18	effect detected ($g = -0.47$, perm. $p = 0.024$)
RQ4	disruption of favourite	12	no effect; third condition unavailable by design
RQ5	classification	1	chance (AUC 0.547, $p = 0.26$)

Table 5: All research questions proposed. RQ3.b was analyzed after the prespecified tests returned null and is exploratory.

5 Discussion

The level of Ψ does not distinguish giant killings from expected results at any interaction order, for any macro feature, and under any sensitivity variant. It does not support classification above chance. The one live result shows that temporally the underdog-minus-favourite emergence gap drifts downward during upsets and upward during controls, replicating across six feature–order cells after FDR correction with medium effect sizes and confirmed by permutation. Since negative Ψ denotes drilled, redundancy-dominated play, the reading is that underdogs consolidate structure as a giant killing game unfolds, while favorites lose structure relatively.

Underdogs who win usually concede possession and passing accuracy while producing more clearances, more long balls and more shots on target which implies a disciplined, compact, direct profile. A team playing that way should look increasingly redundancy-dominated as it settles into a defensive block. What our result adds is that the phenomenon is dynamic so it is not that underdogs who win are more drilled on average, the level tests say that they are not, but that they become so relative to the favourite over the course of the match. It also inverts the framing of the original hypothesis. We looked for a surge of synergy and found a consolidation of structure.

We state that this result is exploratory. It was found after the pre-specified tests returned null. The effective number of independent replications is closer to three than six, and one macro feature carries the opposite sign. It is a hypothesis worth pre-registering, not a finding to build on.

5.1 Why the nulls are not decisive

Three reasons, in increasing order of importance.

Power. Minimum detectable effect is $|g| \approx 0.58$ against an upset arm of 30. An effect of $g = 0.3$ is perfectly plausible for a construct this indirect but would be detected here about one time in three. Two pieces of evidence suggest a small real effect rather than pure noise which are the near-unanimity of sign across 324 specifications, and a placebo-calibrated excess of significant cells at $p_{\text{emp}} = 0.08$. Both point the same way and neither is conclusive.

Definitional redundancy. Most of what Ψ records here is arithmetic common to every team in every match. V_{CoM} is the mean of the micro positions, so every player is mechanically informative about it, and the whole-minus-sum form then double-counts that redundancy ten times over. Consistently, the largest standardized effects across the entire study sit on $V_{\text{clust,vel}}$, the one macro feature that is not an average of its micro inputs, in RQ1 (-0.43), in the window/k sensitivity (-0.48 to -0.72, stable across every setting), and in the RQ5 importance ranking, where the two non-linear features are the only ones with a positive-signed contribution. That pattern is what one would expect if a small real signal existed and were being masked by definitional redundancy wherever the macro feature is a linear aggregate.

The criterion cannot express the hypothesis. This is the substantive point, and table 3 makes it precisely. On the four macro features that carry the bulk of the analysis by magnitude, all of them linear aggregates of the micro state, $\Psi > 0$ occurs in 0.004 % of windows and match-level SRR is zero. The understanding that underdogs show positive emergence in bursts has no range in which to be observed there. A whole-minus-sum statistic is the wrong instrument for detecting synergy in a system where redundancy is this dominant, and that conclusion rests on the definitional redundancy above, which is arithmetic and cannot be preprocessed away.

The separate estimator-saturation problem is more contingent than we first took it to be. 24 % of constituent terms sit at the ceiling, but 88.6 % of those lie inside the clock-artefact void of section 4.1; outside it saturation is 3.8 %. So the instrument is compromised by redundancy in principle and was additionally degraded by a preprocessing defect in practice. The two problems have different remedies — a ΦID synergy measure for the first, a corrected clock for the second — and only the first is a statement about the criterion itself.

The corollary is constructive rather than merely deflationary. The two clustering coefficients do have positive range, a quarter of $V_{\text{clust,vel}}$ windows are synergy-dominated, and they are also where every near-significant effect in this study sits. The evidence points to real indicators being visible on non-linear macro features and invisible on linear ones, which is a testable claim and a concrete design prescription for the next study. We would not now interpret the RQ1 null as evidence that giant killings lack emergent coordination; we would interpret it as evidence that this measurement, on these features, cannot see it either way.

5.2 Relation to prior work

Cheng et al. [4] report strong, clean correlations between causal emergence and possession on the same macro features. Why do we see so much less? Three differences, any of which could be decisive. (i) Data quality. Their tracking is complete 25 Hz optical data while ours is reconstructed from sparse freeze frames with a reported off-camera error of 7.33 m [13]. That error is symmetric and averages out for linear aggregates, but the clustering coefficients are non-linear in the pairwise distances, and those are precisely the features on which we see the largest standardised effects and the least redundancy. (ii) Comparison structure. They compare phases and roles within matches, where the definitional-redundancy floor is common to both sides of the comparison and cancels. We compare across matches, where it does not, and where 98–99 % of the variance is within-match noise. (iii) Estimator. They use a Gaussian plugin while we use KSG. The Gaussian estimator has no finite-sample ceiling of the kind documented in section 4.1 and will not saturate. The non-parametric choice was made to weaken an assumption at the cost of a dynamic-range problem we did not anticipate.

None of this contradicts [4]. It does suggest that the within-match results there do not straightforwardly transfer to between-match outcome prediction, and that the transfer requires both better tracking and a synergy measure that does not double-count redundancy.

5.3 Practical implications

From a study whose primary results are null and whose one positive result is exploratory, we may be reluctant to advise tactics for coaches, however three things can be said honestly, A fuller statement for practitioners accompanies the replication package.

First, there is no evidence for the intuition that upsets are produced by a surge of improvised collective creativity. If it exists, it is smaller than $g \approx 0.58$ or invisible to this measure. The practical implication is negative and worth stating: a coach should not expect a measurable "togetherness" spike to be the mechanism.

Second, the one signal we do see points the other way, toward consolidation. The underdogs who win look, relative to their opponents, increasingly synchronised and predictable as the match goes on. If that survives replication, the actionable form is about trajectory management rather than intensity. The question for an underdog is not how coordinated the team is at kick-off but whether its structure holds and tightens as the match progresses, when fatigue and score pressure act against it. That is trainable in a way "wanting it more" is not.

Third, structure appears to matter more than movement synchrony. Every part of this study where anything approached significance involved a spatial-configuration feature, inverse-distance clustering, or the joint feature containing it, and $V_{\text{clust,vel}}$, velocity similarity, is the one feature that consistently either shows nothing or points the other way. This echoes [4], who found velocity-similarity clustering uncorrelated with possession while positional features correlated strongly. Compactness and spacing appear to be where team-level information lives while running in the same direction at the same time does not appear to be.

5.4 Limitations

1. Sample. 30 upsets against a target of 50–75. Every inferential conclusion in this paper is secondary to n.

2. Reconstructed tracking. 7.33 m mean off-camera error, uncalibrated against the Everett et al. [5] imputer. The non-linear clustering features are the most exposed to this and are also where the largest effects sit.

3. Dataset composition. Eight competition-seasons, overwhelmingly international tournaments. Cup competitions, where giant killings are most common and most culturally salient, are absent.

4. No even-game stratum RQ4 is unanswerable by design.

5. **A match-clock defect.** The absolute clock double-counts the first half, so 29.1 % of analysis windows are linear interpolation across a $\approx$ 42-minute void and report $n_{\text{eff}} = 59$ while containing no observations. They carry 88.6 % of all estimator saturation and pull mean $\Psi^{(1)}$ from -8.4 to -17.0. Every number in this paper was computed under that clock. The gap-trend result survives excluding the affected windows ($g = -0.458$, $p = 0.009$ against -0.468, $p = 0.005$), and the level nulls should be robust because the artefact is near-identical across groups, but the corpus should be re-estimated on the corrected clock before any of these figures is quoted elsewhere.

6. Estimator saturation. Roughly a third of window-level Ψ values on the linear features sit at a floor, though see the preceding item: most of that is the clock defect rather than the estimator's dynamic range at $N \approx 59$.

7. Exploratory positive result. RQ3b was not pre-specified.

8. Analytic degrees of freedom. The within-match summary statistic alone moves p from 0.02 to 0.12 on identical data. We report the full specification curve rather than a choice, but the fact remains that this literature can produce either answer.

5.5 Recommendations

1. **Re-estimate the corpus on the corrected match clock** before anything else. This is mechanical, costs one compute run, and removes an artefact that touches 29 % of the analysis windows. Until it is done, every figure here carries an avoidable caveat.
2. Confirm the gap-trend effect out of sample. Pre-register the slope of the underdog–favourite gap as the primary endpoint, with `V_all` at $\ell = 1$ and the mean as the within-match summary, and test on held-out matches. This is the highest-value scientific next step once the clock is fixed.
3. Replace whole-minus-sum Ψ with the ΦID-based synergy of Rosas et al. §III B [14, 11], which does not double-count redundancy across marginals and can therefore represent positive emergence. This addresses the deepest problem identified here.
4. Lengthen the window to $W = 90$–120 s, or increase k, to lift the estimator ceiling. The $V_{\text{clust,vel}}$ effect reaches nominal significance at $W = 90$ and $k = 10$, which is what one expects if saturation is masking signal.
5. Prefer non-linear macro features $V_{\text{clust,vel}}$ and $V_{\text{clust,dist}}$ have the lowest definitional-redundancy floor and carry the largest standardised effects.
6. Expand the upset arm toward 150 matches, drawing on cup competitions, and add an even-match loss stratum so RQ4 becomes answerable.
7. Cross-validate the tracking reconstruction against the LSTM–GNN imputer of [5] before interpreting any clustering-coefficient result.

6 Conclusion

We set out to test whether giant killings carry an information-theoretic signature of emergent coordination. On the level of the causal-emergence coefficient Ψ, they do not. Of the 54 pre-specified tests, three interaction orders, six macro features, a full sensitivity battery and a classifier all return null values. One exploratory result that is significant is the underdog-minus-favourite emergence gap trends downward over the match in upsets and upward in controls which points toward consolidation of structure rather than a surge of synergy.

The more transferable finding is about the instrument. On 60-second windows of football tracking data, Ψ correlates at $r = 0.9996$ with the negated micro sum. It is dominated by the arithmetic fact that a team's centre of mass is the mean of its players' positions — and that domination is specific: on macro features that are linear aggregates of the micro state, Ψ is positive in 0.004 % of windows and the criterion has no usable range; on the clustering coefficients, which are not, it is positive up to a quarter of the time. Researchers applying causal emergence to team sports should expect this and design around it: use synergy measures that do not double-count redundancy, and prefer macro features that are not linear aggregates of the micro state.

A third lesson is duller and was learned the hard way. Roughly a fifth of the corpus appeared to sit at the estimator's dynamic-range ceiling, and we initially read that as a property of estimating mutual information from 59 samples. It was not: 88.6 % of it traced to a double-counted match clock that filled 29 % of every match with interpolated motion, invisible to all six of our diagnostics because those windows report a full complement of valid frames. A saturating estimator is a symptom worth chasing to its cause before it is written up as a limitation. The question of whether giant killings are emergent remains open. It is open for better reasons now than when we started.

Data and code availability

All code, processed data, figures and result tables are in the replication package accompanying this manuscript. Source data are StatsBomb open data [17], FBref, Transfermarkt and Understat, all publicly available and used in accordance with their terms. No personally identifiable information is used; tracking data are aggregated to team-level features. Raw provider data are not redistributed; the pipeline reproduces them from public sources.

Ethics

This study analyses publicly available match data on professional athletes acting in a public professional capacity. No human-subjects approval was required.

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A Output schema

`data/caus_emg/{match_id}_caus.{csv,parquet}` — one row per (match, team, window, macro feature); 63 columns.

Group	**Columns**
Identity	`match_id, outcome, competition, team, opponent, role, team_rank`
Window	`window_start, t_abs, minute, macro_feature, n_eff, frac_valid`
Information	`I_macro, H_macro, I_micro_l1..3, I_micro_l1c`
Ψ	`psi_l1, psi_l2, psi_l3, psi_l3_raw, psi_l1c`
Subset counts	`n_terms_l1..3, n_total_l1..3`
EAI	`eai_l1..3, eai_stab_l1..3, eai_denom_l1`
DC	`i_cross, h_fav, h_und, dc, dc_rev, dc_ic`
SRR	`srr_l1..3, srr_roll_l1..3, srr_match_l1..3`
Diagnostics	`estimate_ok, neg_mi_flag, mono_violation, psi_l1_z, is_outlier, eai_unstable, dc_unstable`

B Diagnostic counts from the production run

Matches processed	136 (all `ok`)
Window-level records	1,384,566
Distinct windows	116,709
Estimates successfully computed	99.98 %
Median usable frames per window	59 of 59
Reconciliation error against Sprint 3	0.0 nats over 9,371 windows
Compute	4,898 core-minutes
Flagged: MAD outliers	6,049
Flagged: negative I_{macro}	4,255
Flagged: order-monotonicity violations	55,594
Flagged: unstable EAI denominator	58
Flagged: unstable DC denominator	147,456

The large DC flag count is the differential-entropy problem of section 3.5 doing its job: $|H(V_f)| < 1$ nat for the low-variance clustering features, so the literal ratio is uninterpretable there and dc_{ic} should be used instead.